

A K -SPACE DIRECTIVITY SOLVER FOR MULTI-CHANNEL PARAMETRIC ARRAY LOUSPEAKERS

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ABSTRACT

Multi-channel parametric array loudspeakers (PALs) provide enhanced flexibility for directional audio control, but accurate far-field directivity prediction remains challenging due to the nonlinear interactions among multiple ultrasonic fields. The generated audio directivity is determined by the spatial distribution of nonlinear acoustic sources, which depends on ultrasonic propagation and multi-channel excitation. This paper presents a spatial Fourier-domain observable-oriented directivity solver for PALs. The proposed framework evaluates the spatial Fourier transform of the nonlinear source distribution on the audio k -space shell to obtain far-field directivity for different transducer configurations and observation planes. To enable large-scale simulations, an observable-oriented dimension-reduction formulation and a block-wise streaming strategy are introduced to avoid storing the complete three-dimensional nonlinear source distribution. The proposed implementation reduces the peak memory consumption from 5.97 GB to 2.00 GB for a representative large-domain simulation. The proposed solver is benchmarked against modified convolution directivity models, showing good agreement for continuous apertures and improved capability for discrete multi-channel steered PAL arrays.

Index Terms— Parametric array loudspeaker, far-field directivity, Fourier-domain method, multi-channel array, complexity and memory analysis

1. INTRODUCTION

Parametric array loudspeakers (PALs) generate audible sound through the nonlinear interaction of ultrasonic primary waves [1, 2, 3, 4, 5, 6, 7, 8]. Owing to their narrow audio beams and compact ultrasonic apertures, PALs have been studied for directional audio reproduction, private sound zones, and spatial audio applications [3, 9, 10]. Recent multi-channel PAL systems further introduce channel-wise amplitude and phase control, enabling electronic steering and more flexible transmit-side sound-field design [11, 12, 10, 13, 14, 15, 16]. This additional flexibility, however, also makes audio far-field directivity prediction more difficult because the generated field is no longer determined by a simple linear array factor [17, 18].

Unlike conventional loudspeaker arrays, PALs rely on nonlinear self-demodulation, and the audible field is produced by a distributed virtual source whose spatial structure depends on ultrasonic propagation, aperture geometry, element discreteness, and multi-channel excitation [1, 2, 3, 19]. For practical discrete arrays, steering phases and ultrasonic grating lobes can produce complex virtual-source structures that directly affect the resulting audio directivity [17, 18, 19]. A useful simulator should therefore connect the ultrasonic transmit configuration to the far-field audio field through the intermediate virtual-source distribution.

Several analytical and semi-analytical models have been developed for PAL directivity prediction. Convolution-based models relate the audio directivity to the directivities of the ultrasonic primary fields and provide efficient approximations [17, 18]. However, for discrete multi-channel arrays, steering-dependent interference and element layout can produce spatial structures that are difficult to represent using simplified models. A direct numerical formulation that preserves the virtual-source distribution is therefore desirable.

A k -space formulation provides a natural framework for representing the nonlinear virtual source and evaluating its far-field radiation [20, 21, 22, 19]. Based on this relation, we present an observable-oriented k -space directivity solver for multi-channel PALs. The solver computes the ultrasonic primary fields, constructs the nonlinear virtual source, and evaluates its Fourier-domain contribution only along prescribed observation directions. To improve scalability, an observation-plane reduction and a block-wise streaming strategy are introduced to avoid storing the complete three-dimensional virtual source. The method is validated through simulation-domain convergence, computational-cost measurements, and comparison with the modified convolution directivity model for both continuous apertures and discrete steered arrays.

2. THEORY

2.1. k -Space directivity formulation

Let the two ultrasonic primary pressures be $p_1(\mathbf{r}, \omega_1)$ and $p_2(\mathbf{r}, \omega_2)$. The generated audio difference frequency is $\omega_a = \omega_2 - \omega_1$, with $k_a = \omega_a/c_0$. Under the quasi-linear approxi-

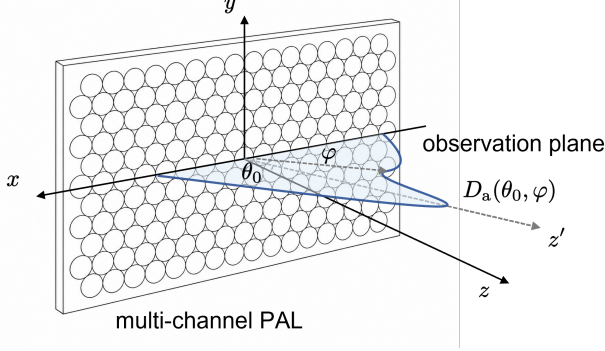


Fig. 1. Geometry of the multi-channel PAL aperture and the prescribed observation plane for evaluating $D_a(\theta_0, \varphi)$.

mation, the equivalent audio virtual source is

$$q_a(\mathbf{r}, \omega_a) = \frac{\beta \omega_a^2}{\rho_0 c_0^4} p_1^*(\mathbf{r}, \omega_1) p_2(\mathbf{r}, \omega_2). \quad (1)$$

The audio pressure generated by the virtual source can be expressed using the free-space Green's function as

$$p_a(\mathbf{r}, \omega_a) = -i \rho_0 \omega_a \iiint q_a(\mathbf{r}_v, \omega_a) \frac{e^{ik_a |\mathbf{r} - \mathbf{r}_v|}}{4\pi |\mathbf{r} - \mathbf{r}_v|} d^3 \mathbf{r}_v. \quad (2)$$

For far-field observation, $|\mathbf{r} - \mathbf{r}_v| \approx r - \hat{\mathbf{s}} \cdot \mathbf{r}_v$, where $\hat{\mathbf{s}} = \mathbf{r}/r$. The pressure then becomes

$$p_a(\mathbf{r}, \omega_a) = \frac{e^{ik_a r}}{r} D_a(\hat{\mathbf{s}}, \omega_a), \quad (3)$$

where

$$D_a(\hat{\mathbf{s}}, \omega_a) = -\frac{i \rho_0 \omega_a}{4\pi} \iiint q_a(\mathbf{r}_v, \omega_a) e^{-ik_a \hat{\mathbf{s}} \cdot \mathbf{r}_v} d^3 \mathbf{r}_v. \quad (4)$$

Equation (4) shows that the far-field directivity is determined by the spatial spectrum of the virtual source evaluated on the audio k -space shell,

$$D_a(\hat{\mathbf{s}}, \omega_a) = -\frac{i \rho_0 \omega_a}{4\pi} \tilde{q}_a(\mathbf{k}, \omega_a) \Big|_{\mathbf{k} = k_a \hat{\mathbf{s}}}, \quad (5)$$

where

$$\tilde{q}_a(\mathbf{k}, \omega_a) = \iiint q_a(\mathbf{r}_v, \omega_a) e^{-i\mathbf{k} \cdot \mathbf{r}_v} d^3 \mathbf{r}_v. \quad (6)$$

This formulation connects the PAL transmit configuration with the far-field audio directivity through the intermediate virtual-source distribution. In the proposed solver, the primary ultrasonic fields are calculated first, the virtual source is then constructed according to Eq. (1), and the far-field directivity is obtained by evaluating the spatial spectrum of the virtual source on the audio k -space shell.

2.2. Observable-oriented reduction and streaming

As shown in Fig. 1, the far-field directivity is evaluated only in a prescribed observation plane specified by θ_0 . Instead of computing the directivity over the full sphere, the proposed solver focuses on the observation directions within this plane. This observable-oriented setting allows the virtual-source distribution to be reduced before the final spatial-spectrum evaluation. The observation direction is parameterized as

$$\hat{\mathbf{s}}(\theta_0, \varphi) = (\sin \varphi \cos \theta_0, \sin \varphi \sin \theta_0, \cos \varphi), \quad (7)$$

where φ is the scanning angle inside the prescribed observation plane.

Taking the xOz plane as a representative case, namely $\theta_0 = 0$, the observation direction becomes $\hat{\mathbf{s}}(0, \varphi) = (\sin \varphi, 0, \cos \varphi)$. Since the observation direction has no y component, the virtual source can be projected along y as

$$q_{xz}(x, z, \omega_a) = \int_{-\infty}^{\infty} q_a(x, y, z, \omega_a) dy. \quad (8)$$

The far-field directivity in this observation plane can then be evaluated from the reduced source distribution rather than from the complete three-dimensional virtual source. The same principle can be applied to other prescribed observation planes by selecting the corresponding projection direction.

The complete virtual-source domain required for PAL directivity convergence can be very large. Therefore, as illustrated in Fig. 2, the proposed implementation adopts a block-wise streaming strategy along the propagation direction. For the i -th axial block, the primary fields are calculated, the local virtual source is generated, and its contribution to the far-field directivity is accumulated,

$$p_1^{(i)}, p_2^{(i)} \rightarrow q_a^{(i)} \rightarrow D_a^{(i)}. \quad (9)$$

The final directivity is obtained by summing the contributions from all blocks,

$$D_a(\theta_0, \varphi, \omega_a) = \sum_{i=1}^{N_b} D_a^{(i)}(\theta_0, \varphi, \omega_a). \quad (10)$$

During this process, only the current block and the accumulated far-field contribution are stored. The proposed observable-oriented reduction and block-wise streaming strategy therefore avoids storing the complete three-dimensional virtual source while preserving the physical link between ultrasonic propagation, virtual-source generation, and far-field audio directivity.

3. NUMERICAL VALIDATION AND COMPUTATIONAL PERFORMANCE

3.1. Simulation-domain convergence

The virtual-source computational domain is determined using a large rectangular PAL aperture with $W = 0.165$ m and $H =$

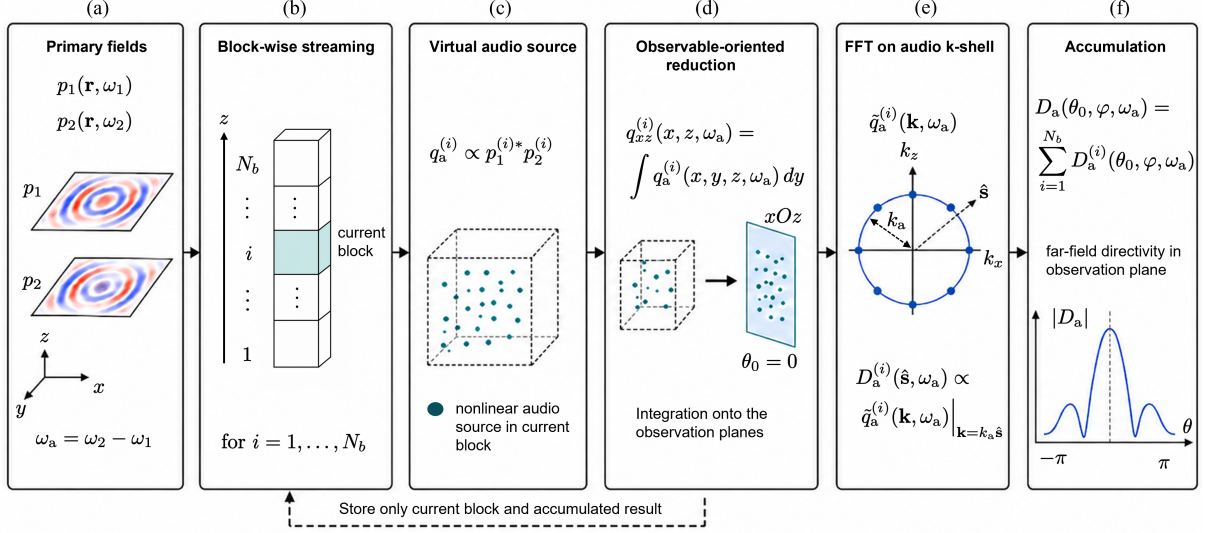


Fig. 2. Workflow of the proposed solver: block-wise ultrasonic propagation, virtual-source generation, observation-plane reduction, and k -space far-field evaluation.

0.109 m. The primary frequencies are $f_1 = 40$ kHz and $f_2 = 41$ kHz, giving $f_a = 1$ kHz. The domain is defined as $x \in [-L_x/2, L_x/2]$, $y \in [-L_y/2, L_y/2]$, and $z \in [0, L_z]$, with $\Delta x = \Delta y = 0.01$ m and $\Delta z = 0.005$ m. In Fig. 3, $z_{\max}$ denotes the same axial length as L_z , and the transverse scan uses $L_x = L_y = L_{xy}$.

Convergence is evaluated by comparing the normalized directivity with the largest-domain result. The relative levels at $\varphi = 20^\circ, 40^\circ$, and 60° are used as metrics. As shown in Fig. 3(a) and (b), the selected-angle differences fall below 0.3 dB when $L_z \geq 24$ m. Therefore, $L_z = 24$ m is used in the following simulations. With this axial length fixed, Fig. 3(c) and (d) show that the transverse domain converges at approximately $L_{xy} = 1.4$ m. For steered sources, the transverse domain should be further expanded along the steering direction to cover the tilted virtual-source region.

3.2. Computational efficiency improvement

The proposed solver improves scalability through observable-oriented dimension reduction and block-wise streaming. The full k -space implementation stores the complete three-dimensional virtual source, whereas the observable-oriented implementation processes the virtual source block by block and accumulates only the contribution required for the prescribed observation plane.

The theoretical complexity and memory requirements are summarized in Table 1, where N_φ is the number of observation directions and $N_{z,b} = 64$ is the axial block size used in the streaming implementation.

Table 2 reports measured runtime and peak memory for representative domains. The runtime includes ultrasonic

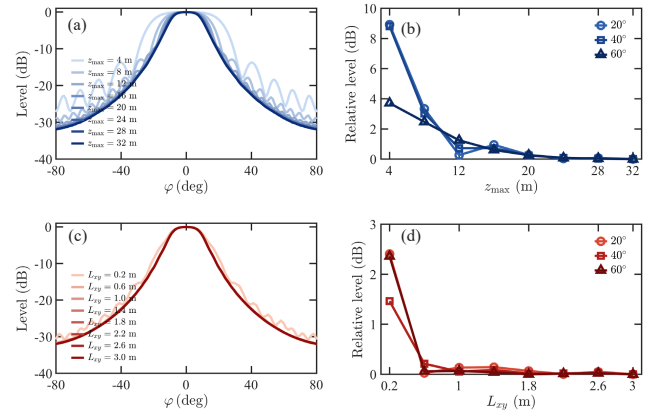


Fig. 3. Simulation-domain convergence for the large rectangular PAL aperture. (a) Directivity curves with different axial lengths $z_{\max}$ at fixed $L_{xy} = 2$ m. (b) Relative levels at selected $z_{\max}$ angles versus $z_{\max}$. (c) Directivity curves with different transverse sizes L_{xy} at fixed $L_z = 24$ m. (d) Relative levels at selected observation angles versus L_{xy} .

propagation, virtual-source generation, observation-plane accumulation, and far-field evaluation.

The observable-oriented implementation reduces the runtime from 136.92 s to 42.69 s and the peak memory from 5.97 GB to 2.00 GB for $L_z = 24$ m and $L_{xy} = 2$ m. For the larger transverse domain, the full k -space implementation becomes infeasible, while the observable-oriented solver remains executable with 2.44 GB peak memory.

Table 1. Computational complexity and memory requirement.

Method	Complexity	Memory
Full k -space	$O(N_x N_y N_z N_\varphi)$	$O(N_x N_y N_z)$
Observable-oriented k -space	$O(N_x N_y N_z)$ + $O(N_x N_z N_\varphi)$	$O(N_x N_y N_z)$ + $O(N_x N_z)$

Table 2. Measured computational cost of the full and observable-oriented k -space implementations.

Method	L_z (m)	L_{xy} (m)	Runtime (s)	Peak memory (GB)
Observable-oriented k -space	24	2	42.69	2.00
Observable-oriented k -space	36	2	63.91	2.06
Observable-oriented k -space	24	4	167.13	2.44
Full k -space	24	2	136.92	5.97
Full k -space	24	4	—	> 45.8 [†]

[†]The full k -space implementation failed because an internal array larger than 45.8 GB was requested.

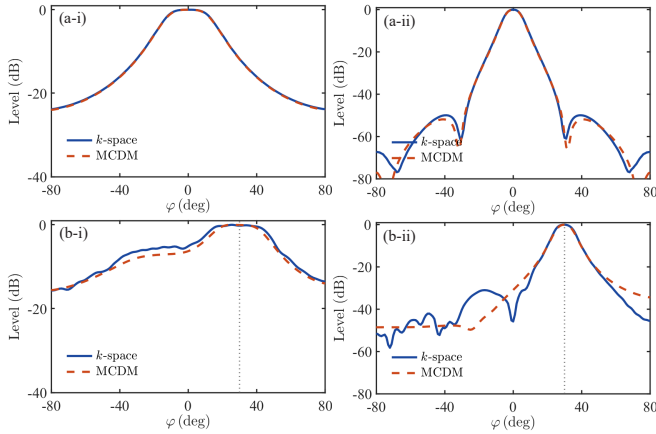


Fig. 4. Comparison between the observable-oriented k -space solver and MCDM at different difference frequencies. (a-i) Continuous aperture at $f_a = 500$ Hz. (a-ii) Continuous aperture at $f_a = 4$ kHz. (b-i) 32-channel staggered array at $f_a = 500$ Hz. (b-ii) The same array at $f_a = 4$ kHz.

3.3. Comparison with MCDM

The observable-oriented k -space solver is compared with the modified convolution directivity model (MCDM), using the k -space result as the numerical reference. Two sources are tested: a non-steered continuous circular aperture with radius $a = 0.10$ m, and a 32-channel staggered PAL array steered to $\theta_s = 30^\circ$, where each channel drives one column of six circular elements with radius $a = 5$ mm. The continuous case uses $L_z = 30$ m and $L_x = L_y = 2.4$ m, while the steered array uses $L_z = 36$ m, $L_x = 36$ m, and $L_y = 0.96$ m.

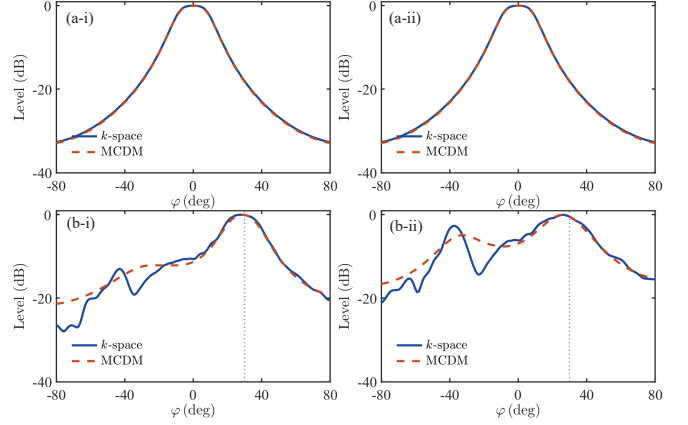


Fig. 5. Comparison between the observable-oriented k -space solver and MCDM on different observation planes at $f_a = 1$ kHz. (a-i) Continuous aperture at $\theta_0 = 15^\circ$. (a-ii) Continuous aperture at $\theta_0 = 30^\circ$. (b-i) 32-channel staggered array at $\theta_0 = 15^\circ$. (b-ii) The same array at $\theta_0 = 30^\circ$.

Figure 4 compares the two methods in the main observation plane. For the continuous aperture, the maximum errors above -20 dB are below 0.5 dB at both $f_a = 500$ Hz and 4 kHz. For the discrete steered array, MCDM shows larger deviations, especially in the sidelobe and null regions.

Figure 5 further evaluates off-plane observations. The continuous aperture remains consistent with MCDM, with errors below 0.2 dB above -20 dB. In contrast, the discrete array exhibits errors of several dB, indicating that MCDM can miss observation-plane-dependent sidelobe structures produced by practical multi-channel excitation.

4. CONCLUSION

This paper presented an observable-oriented k -space directivity solver for parametric array loudspeakers. By constructing the nonlinear audio virtual source from ultrasonic primary fields and evaluating its Fourier projection on the audio k -shell, the proposed method directly links transducer excitation to far-field audio directivity on prescribed observation planes. The observation-plane reduction and block-wise streaming strategy reduce the memory required for large-domain simulations while preserving the spatial structure of the virtual source. Numerical results show close agreement with MCDM for smooth continuous apertures and clear advantages for practical discrete steered arrays, where sidelobe and null structures are strongly affected by element discreteness, channelization, and steering.

5. COMPLIANCE WITH ETHICAL STANDARDS

No ethical approval was required for this numerical simulation study. The authors have no relevant financial or non-financial interests to disclose.

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